

Reducing Churn with Machine Learning

A data-driven retention program for Company A (Telecom)

GCI WORLD 2026 — FINAL ASSIGNMENT

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Our Approach: The Final Assignment Workflow

We follow the **5-step iterative process**, combining a Data Engineering Approach with a Business Mindset at every stage. The loop repeats until the proposal is validated and ready to scale.



Data Engineering Lens

Every step is grounded in rigorous data exploration, feature engineering, and model validation — ensuring technical soundness before any recommendation is made.

Business Mindset Lens

Every analytical finding is translated into a concrete, actionable business decision — identifying bottlenecks, stakeholders, and measurable outcomes at each stage.

STEP 1 — UNDERSTAND THE BACKGROUND

The Business Context: A Mature, Competitive Market

Key Message: Telecom is a mature, saturated market where retaining customers is far cheaper than acquiring them.

Who Is Company A?

A wireless telecom operator with **~100,000 customers** in a saturated market. Switching is easy, price competition is intense, and differentiation is limited.

Common Churn Drivers

- Poor network quality or coverage
- Better competitor pricing
- Outdated devices
- Weak customer service

Why Retention Is the Priority

Churn Benchmarks

- Annual churn in telecom ranges ~20–50% depending on market and segment [3][4]; prepaid up to ~70%, postpaid ~10–20% [3].
- Regional benchmarks: North America ~14.6–20%, Europe ~20–44%, Asia-Pacific ~30–50% [3]. Verizon reported 1.22% monthly churn (~14.6%/yr); T-Mobile US prepaid 2.75% in Q1 2024 [3][5].

Retention Economics

- Acquiring a new customer costs 5–25× more than retaining one [1]; in telecom 6–7× (~$300 per subscriber) [3].
- Increasing retention by just 5% can raise profits 25–95% (Reichheld / Bain & Company) [1][2].

5–25×

Cost of acquisition vs. retention

100K

Customers in the base

STEP 1 — UNDERSTAND THE BACKGROUND

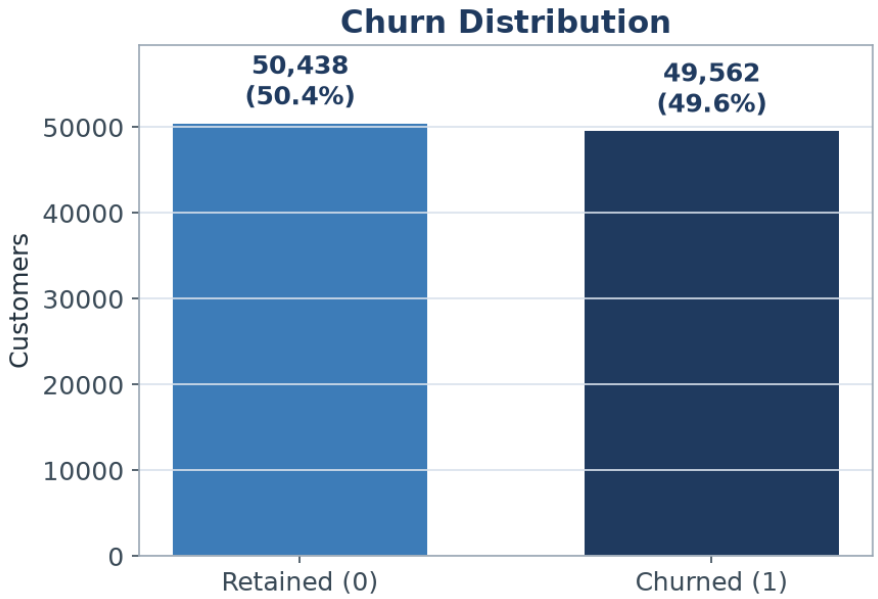
Exploratory Data Analysis: The Numbers Behind the Problem

Key Message: EDA and data profiling revealed a **49.6% churn rate** — translating to ~$34.4M in annual revenue at risk.

Dataset Overview

- **Source:** Client.csv + Record.csv
- **Customers:** 100,000 rows
- **Features:** ~90 variables
- **Target:** Churn (0 = stays, 1 = leaves)

Key features include minutes of use, overage minutes, device age, tenure, handset price, and revenue indicators.



49,562 customers churned — a near 50/50 split, indicating a severe and urgent retention problem. Annual revenue at risk: **~$34.4M**.

STEP 2 — CREATE A HYPOTHESIS

From EDA to a Testable Hypothesis

Key Message: Churners share a clear, consistent profile — we translate that into a **testable, actionable hypothesis.**

What the EDA Revealed

Customers who churn consistently show three characteristics:

Higher overage minutes

Paying for usage beyond their plan

Older devices

Longer time since last equipment upgrade

Shorter tenure

Less loyalty, more price-sensitive

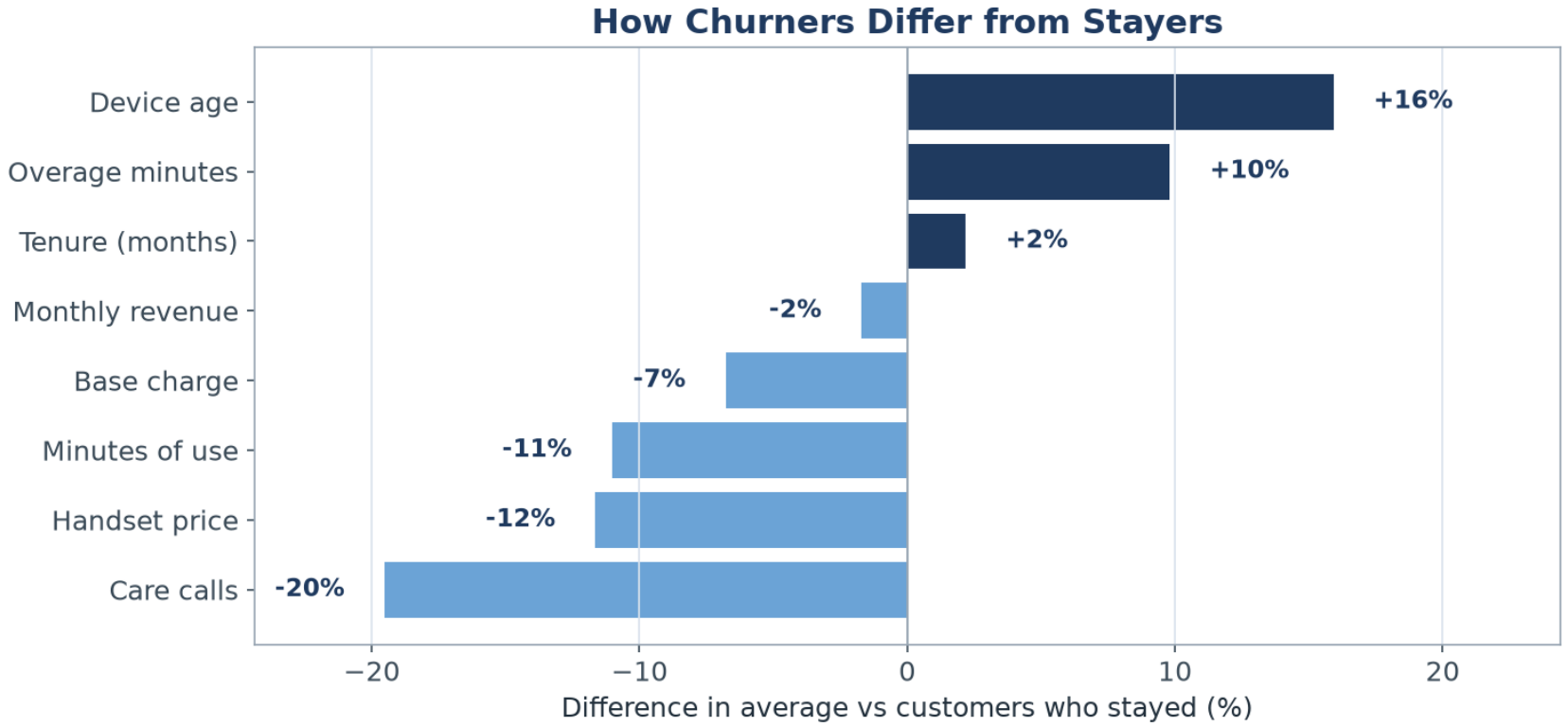
Our Hypothesis

Outdated device + overage charges + short tenure = high churn risk.

Each driver is **actionable** — a subsidized device, a plan adjustment, or a loyalty offer can directly address the root cause.

ML Task Definition

Supervised **binary classification** on the churn target, using the full feature set to predict individual churn probability.


How Churners Differ from Stayers
Device age
Overage minutes
Tenure (months)
Monthly revenue
Base charge
Minutes of use
Handset price
Care calls
+16%
+10%
+2%
-2%
-7%
-11%
-12%
-20%
−20
−10
0
10
20
Difference in average vs customers who stayed (%)

STEP 3 — DEEPEN YOUR HYPOTHESIS

Feature Engineering, Model Training & Bottleneck Identification

Key Message: XGBoost confirms the hypothesis — **device age, refurbished status, and tenure** are the dominant churn drivers.

Data Engineering Approach

- Focus analysis on top EDA signals
- Create new features: **overage_ratio, rev_per_mou, rev_declining**
- Train XGBoost model to quantify driver importance

Business Mindset: Identify the Bottleneck

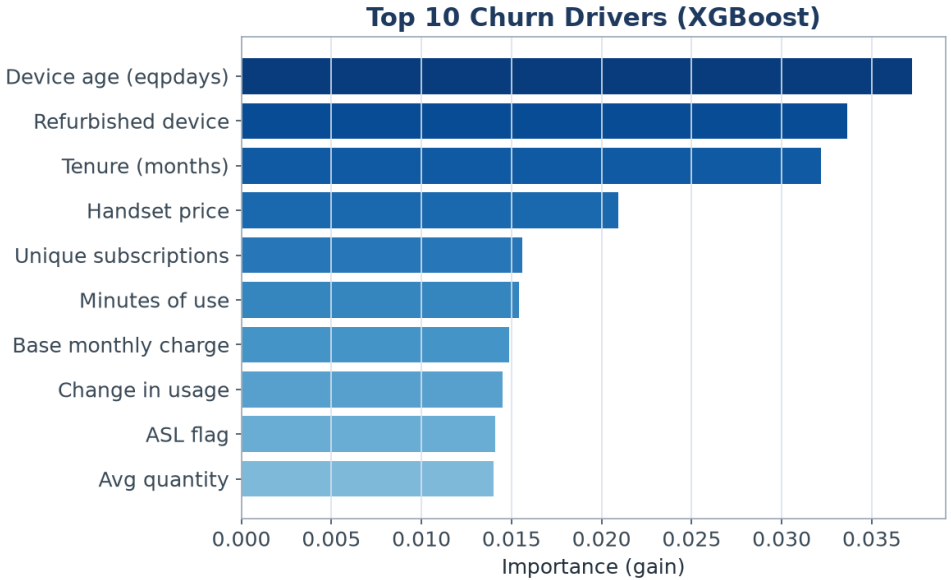
Bottleneck: Customers are leaving because they feel underserved — paying overage charges while using outdated hardware, with no proactive outreach from Company A.

Stakeholders

- Customer retention team
- Device supply & logistics
- Marketing & pricing

Current vs. Proposed

Today: Reactive call-center retention after a churn signal.
Proposed: Proactive, predictive outreach *before* the customer decides to leave.



STEP 4 — CREATE YOUR PROPOSAL

Targeting the Right Customers: High-Risk + High-Value

Key Message: We target **13,574 customers** with churn probability >60% AND above-median revenue — where $1.19M monthly revenue is concentrated.

Segment Definition

Churn Probability > 60%

Model-flagged high-risk customers

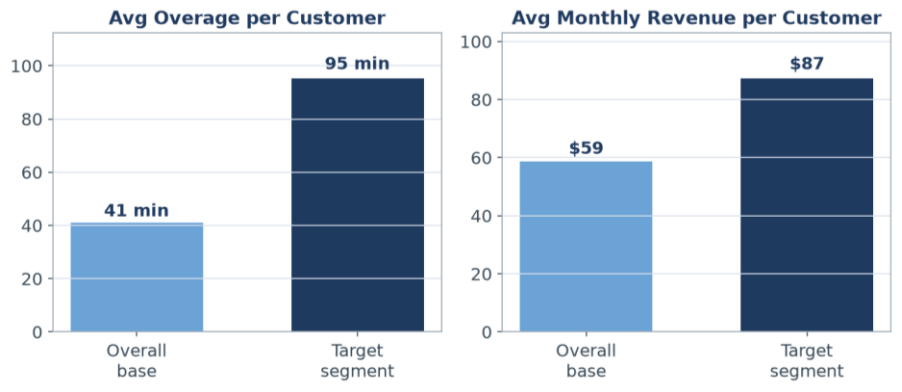
Above-Median Revenue

High-value customers worth saving

13,574 Customers

The intersection of both criteria

Target Segment vs. Overall Base

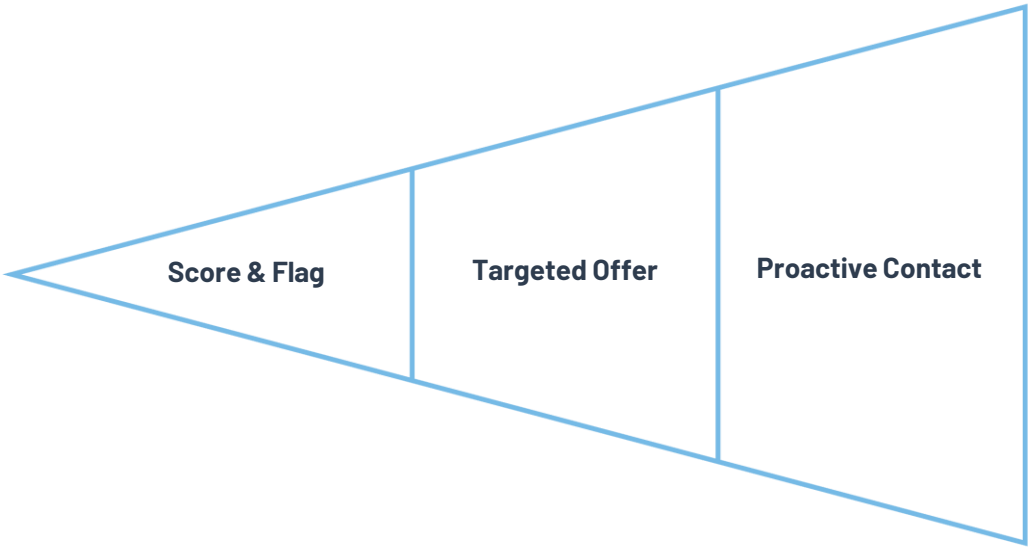


The target segment has **132% higher overage** (95 vs 41 min) and concentrates **$1.19M in monthly revenue** — making intervention highly impactful.

STEP 4 — CREATE YOUR PROPOSAL

The Retention Program: A Monthly AI-Driven Pipeline

Key Message: A repeatable monthly pipeline that detects and contacts at-risk customers **before** they churn — with personalized offers mapped to each driver.



Each offer maps directly to a driver the model identified:

Subsidized Device Upgrade
Addresses outdated hardware (eqpdays, refurbished flag)

Plan Adjustment
Eliminates overage charges — the strongest behavioral signal

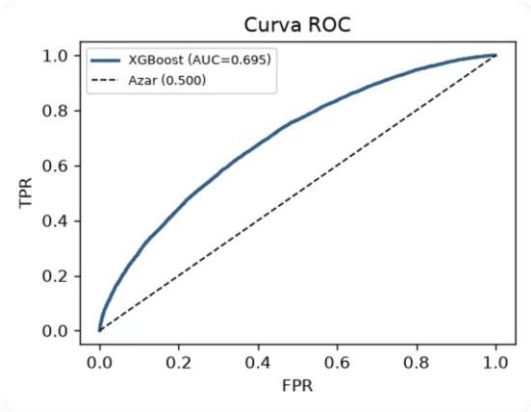
Loyalty Discount
Rewards tenure and reduces price sensitivity for short-tenure customers

STEP 5 — VERIFY YOUR PROPOSAL

Model Performance & Business Impact

Key Message: The model works and the business case is compelling — **net benefit of $7.89M with an 11.6× ROI.**

Model Evaluation (XGBoost)



0.695

ROC-AUC

Discriminatory power

63.8%

Accuracy

Overall correctness

0.64

Recall (Churn)

Churners correctly caught

Business Impact Summary

13,574 Contacted

High-risk + high-value segment

4,072 Retained

Conservative 30% conversion rate

$8.57M Revenue Saved

Value over 24 months (LTV)

$7.89M Net Benefit

After $678,700 campaign cost — **11.6× ROI**

A **device-based offer outperforms** a blanket discount — validating the driver-mapped approach.

Made with GAMMA

REPEAT THE PROCESS

Iterate, Validate, and Scale

Key Message: We iterate on results — running controlled experiments, enriching data, and improving the model before scaling.

Next Steps to Validate & Scale

1 Run an A/B Test

Treatment vs. control group to measure the real causal effect of each offer type.

2 Retrain Monthly

Refresh the model with new data; enrich with call-center transcripts (NLP) for deeper signals.

3 Human-in-the-Loop

Moderate AUC (0.695) means the model guides decisions — agents make the final call.

References

1. Gallo, A. (2014). The Value of Keeping the Right Customers. Harvard Business Review. hbr.org/2014/10/the-value-of-keeping-the-right-customers
2. Reichheld, F. & Sasser, W.E. (1990). Zero Defections: Quality Comes to Services. Harvard Business Review. hbr.org/1990/09/zero-defections-quality-comes-to-services
3. Tridens Technology. Why Telecom Customers Churn and How to Measure It. tridenstechnology.com/telecom-churn
4. CustomerGauge. Average Churn Rate by Industry (2025 B2B Benchmarks). customergauge.com/blog/average-churn-rate-by-industry
5. Statista. T-Mobile US customer churn rate by quarter. statista.com/statistics/219793
6. Sagacity Solutions. Customer Retention vs. Acquisition Costs.
7. XGBoost documentation. xgboost.readthedocs.io
8. scikit-learn documentation. scikit-learn.org
9. GCI World 2026 — Matsuo-Iwasawa Laboratory, The University of Tokyo (lecture slides).
10. GAMMA IA to make this presentation.

This proposal follows the iterative methodology — from background understanding through verified, ROI-positive recommendation.